

IRCTC Rail Connect – UX Audit & Redesign

Internship Task 1 – UI Detective

- UX Audit
- High-Fidelity Redesign
- Interactive Prototype

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1. Task Objective

The objective of this task was to evaluate the IRCTC Rail Connect user interface from a usability perspective, identify key user-experience issues, and propose a high-fidelity redesign that improves clarity, consistency, navigation, and overall ease of use.

The redesign focused on the core railway ticket-booking journey, from searching for a train through passenger details, review, and payment. The identified usability issues were translated into practical interface improvements and an interactive prototype.

2. UX Audit Findings

The UI audit identified 15 usability issues across the IRCTC Rail Connect booking experience. The issues were evaluated using established usability principles and were documented with supporting screenshots and observations. The findings mainly related to information hierarchy, navigation, consistency, accessibility, cognitive load, and clarity of user actions.

2.1 Summary of Identified Issues

ID	Usability Issue	Severity
UX-01	Home Screen – Equal visual importance given to multiple services	Major(3)
UX-02	Home Screen – Underutilized space for useful travel information	Moderate(2)
UX-03	Home Screen – Important actions lack clear task-oriented grouping	Moderate(2)
UX-04	Passenger Details - Excessive information and optional decisions on one screen	Major(3)
UX-05	Passenger Details – No visible booking progress indicator	Major(3)
UX-06	Search Results – Several controls are crowded and unclear	Moderate(2)
UX-07	Search Results – High information density in train cards	Major(3)
UX-08	Search Results – Availability relies heavily on color	Moderate(2)

UX-09	Search Results – Difficult comparison between train options	Moderate(2)
UX-10	Search Results - Some icons have unclear meaning	Moderate(2)
UX-11	Loading – Limited feedback during loading	Moderate(2)
UX-12	Login – Multiple competing actions	Major(3)
UX-13	Account Menu – Poor grouping of account functions	Moderate(2)
UX-14	More Section – Poor grouping of secondary features	Major(3)
UX-15	Booking – Travel insurance is preselected	Major(3)
UX-16	Login – Login filed/authentication clarity	Moderate(2)
UX-17	Search Form – Requires multiple selections before searching	Moderate(2)
UX-18	Search Form- Required and optional choices aren't clearly distinguished	Moderate(2)
UX-19	Transactions – Users may need to explore categories to find actions	Moderate(2)
UX-20	Exit Confirmation – Additional interaction can create friction	Minor(1)

The audit revealed recurring usability concerns around information overload, weak content hierarchy, unclear grouping, excessive choices, limited system feedback, accessibility, and insufficient visibility of the booking process. These findings were used to prioritize improvements for the redesigned booking journey.

3. Redesign Approach

The redesign was guided by the usability issues identified during the audit. Rather than changing the interface purely for visual improvement, the redesign focused on addressing specific usability concerns through clearer information hierarchy, simplified interactions, consistent layouts, progressive disclosure, and improved visibility of the booking process.

The redesign focused on the core ticket-booking journey and translated the audit recommendations into a five-screen high-fidelity prototype:

Home → Search Results → Passenger Details → Review → Payment

3.1 Key Design Principles

1. **Simplify the Primary Journey**
The train-booking journey was given stronger priority while secondary services and less frequently used options were visually reduced or grouped.
2. **Improve Information Hierarchy**
Important information such as train timings, journey duration, availability, fare, and primary actions was given clearer visual priority.
3. **Reduce Cognitive Load**
Complex screens were divided into logical sections, while optional preferences were separated from essential information.
4. **Improve System Visibility**
A five-step booking progress indicator was introduced to show the user's position in the journey from Search to Pay.
5. **Improve Clarity and Accessibility**
Text labels were used alongside controls and availability states were communicated through text rather than relying on color alone.

4. High-Fidelity Redesign

Based on the findings from the UX audit, a high-fidelity redesign was developed for the core IRCTC Rail Connect booking journey. The redesign consists of five screens that guide the user from train search through payment while maintaining a consistent visual language and interaction flow.

4.1 Home Screen

The redesigned Home screen gives stronger visual priority to train booking, which is the primary task for most users. Secondary travel services are visually separated to reduce clutter, while the navigation structure is organized around clearer task-oriented categories.

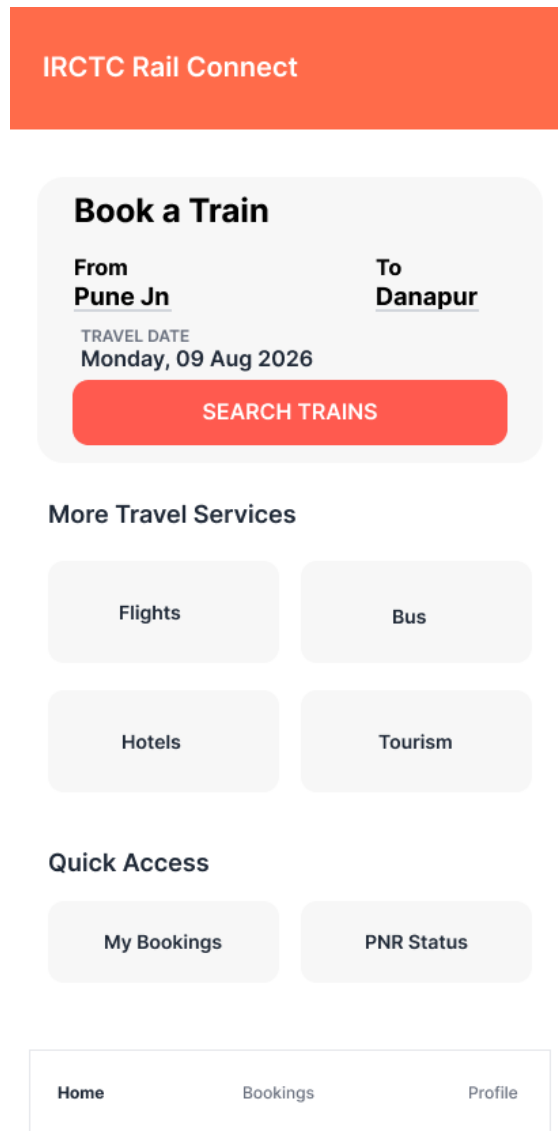


Figure 1. Redesigned IRCTC Home Screen

4.2 Search Results

The redesigned Search Results screen improves train comparison by creating a clearer hierarchy for essential information such as departure and arrival times, journey duration, availability, and fare. Search and filtering controls are organized more clearly, while less important details are reduced to make comparing train options easier

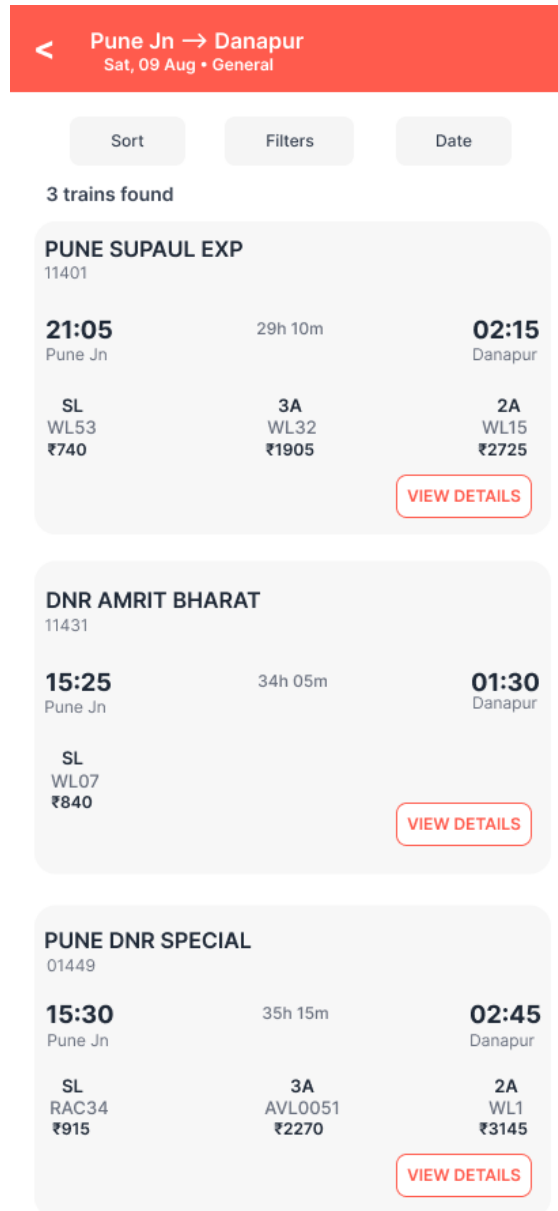


Figure 2. Redesigned IRCTC Search Results Screen

4.3 Passenger Details

The redesigned Passenger Details screen separates essential passenger information from optional preferences. A five-step progress indicator — Search, Select, Passenger, Review, and Pay — makes the user's position in the booking journey visible. Optional settings such as berth preference are kept separate from the core passenger information to reduce cognitive load and make the form easier to complete.

<

Passenger Details

Search

→

Select

→

Passenger

→

Review

→

Pay

PUNE DNR SPECIAL

01449

Pune Jn → Danapur

15:30

02:45

Passenger Information

1 Adult

Full Name

Bhavya

Age

21

Gender

Female

Travel Preference

Berth Preference

No Preference

More Preferences

CONTINUE TO REVIEW

Figure 3. Redesigned IRCTC Passenger Details Screen

4.4 Review

The redesigned Review screen provides a concise summary of the selected journey, passenger information, travel preference, and fare before payment. Information is presented in clearly separated sections so users can verify their booking without navigating back through the entire process. The travel-insurance choice is also presented explicitly rather than being preselected.

<

Review Journey

Search

→

Select

→

Passenger

→

Review

→

Pay

PUNE DNR SPECIAL

01449

Pune Jn → Danapur

09 Aug 2026

15:30

02:45

1. Bhavya

Age : 21

Gender : Female

Travel Preference

Berth Preference

No Preference

Fare Summary

Base Fare

₹2305.85

Total Fare

₹2305.85

Travel Insurance

☐ Yes, add travel insurance - 0.45/person

☒ No, I don't want travel insurance

PROCEED TO PAY

Figure 4. Redesigned IRCTC Review Screen

4.5 Payment

The redesigned Payment screen simplifies the final stage of the booking journey by presenting the payment summary and available payment methods in a clear structure. The total payable amount is given strong visual emphasis, while the selected payment method is clearly indicated. A single primary action, **PAY ₹2305.85**, provides a clear next step for completing the booking.

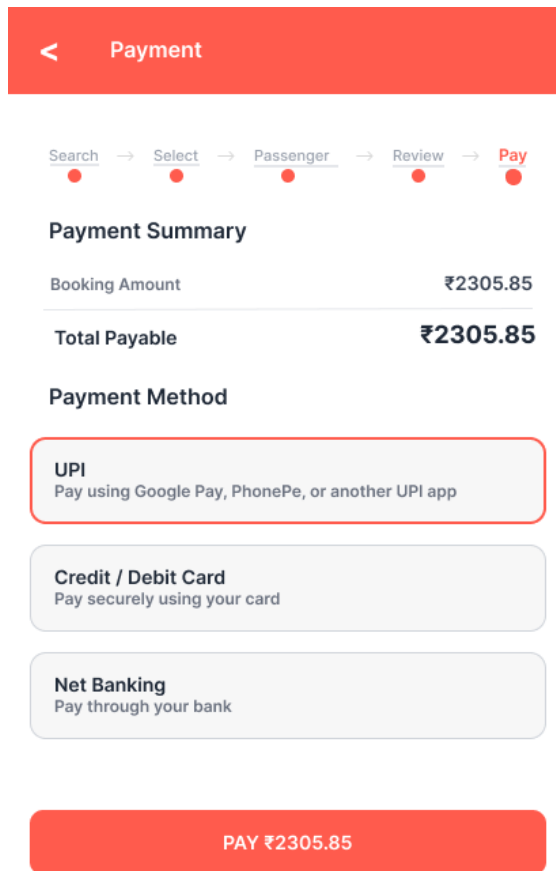


Figure 5. Redesigned IRCTC Payment Screen

5. Prototype Flow

The redesigned booking journey was implemented as an interactive Figma prototype to demonstrate the intended navigation and user flow. The primary journey follows five stages:

Home → Search Results → Passenger Details → Review → Payment

Back-navigation was also implemented between consecutive screens, allowing users to return to the previous stage when required. The complete prototype flow was tested to verify that the primary forward and backward navigation worked as intended.

Figma Prototype:

<https://www.figma.com/proto/5TQdQGk9TLDLITZUMabXMO/IRCTC-Rail-Connect---UX-Design?node-id=6-3&p=f&t=DEISbBhwZovPbJhN-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=6%3A3>

5.1 Prototype Flow Diagram

HOME



SEARCH RESULTS



PASSENGER DETAILS



REVIEW



PAYMENT

6. Key UX Improvements

Audit Finding	Redesign Response
Information overload on key screens	Grouped essential and optional information into clear sections
No visible booking progress	Added a 5-step booking progress indicator
High information density in train results	Created a clearer hierarchy for timings, duration, availability and fare
Difficult comparison between trains	Standardized the information structure across train cards
Unclear search/filter controls	Organized controls into clearer, labeled actions
Reliance on color for availability	Combined visual styling with explicit availability labels

Poor grouping of secondary features	Used clearer sections and task-oriented organization
Travel insurance preselected	Changed it into an explicit user choice
Multiple competing actions	Established one clear primary CTA on each major screen

Overall, the redesign focuses on reducing cognitive load, improving information hierarchy, making system status more visible, and giving users clearer control over their booking decisions. The changes were derived from the findings of the UX audit rather than being purely visual modifications.

7. Conclusion

The IRCTC Rail Connect UX audit identified 20 usability issues across the mobile interface and booking journey. The high-fidelity redesign translated the key audit findings into practical interface improvements focused on clarity, consistency, reduced cognitive load, and improved user control.

The redesigned five-screen booking flow — Home, Search Results, Passenger Details, Review, and Payment — provides a more structured journey from train discovery to payment. An interactive Figma prototype was also created and tested to demonstrate the proposed navigation and interactions.

This redesign demonstrates how usability evaluation can be used to identify user-experience problems and guide meaningful design improvements rather than focusing only on visual changes.